

$$1. (a) P(A|S) = \frac{P(ANS)}{P(S)} = \frac{3}{51} = \frac{1}{17}$$

If we removed a 6, there are only three 6s left

$$(b) P(A|N) = 4/51$$

$$(c) P(A) = 4/52 = \frac{1}{13}$$

(d) D and A are not independent. Regardless of whether the removed card is a red or black 6, there will be less cards to draw from and $\therefore$ a different probability

$$(e) P(D) = \frac{2}{52} = \frac{1}{26}$$

$$2. P(K) = 0.7$$

$$P(K^c) = 0.3$$

$$P(R|K^c) = 1/N$$

$$P(R) = \underbrace{P(R|K)}_{0.7} P(K) + \underbrace{P(R|K^c)}_{0.3/N} P(K^c)$$

$$(a) P(K) = 0.7$$

$$(b) P(R|K) = 1$$

$$(c) P(K|R) = \frac{P(R|K)P(K)}{P(R)} \quad (\text{Bayes Theorem})$$

$$\frac{\frac{P(R|K)P(K)}{P(R|K)P(K) + P(R|K^c)P(K^c)}}{\frac{0.7}{0.7 + \frac{0.3}{N}}}$$

$$(d) P(K|R) > 0.99$$

$$\frac{0.7}{0.7 + \frac{0.3}{N}} > 0.99$$

$$0.7 > 0.693 + \frac{0.297}{N}$$

$$0.007 > 0.297 / N$$

$$0.007N > 0.297$$

$$N > 42.4...$$

$$N \geq 43$$

$$3. P(A) = 0.3$$

$$P(B|A) = 0.75$$

$$P(B|A^c) = 0.2$$

$$P(G|A \cap B) = 0.2$$

$$P(G|A^c \cap B) = 0.15$$

$$P(G|A \cap B^c) = 0.8$$

$$P(G|A^c \cap B^c) = 0.9$$

I'm going to use
G instead of C to differentiate
from parenthesis

$$(a) P(A \cap B \cap G) \quad \text{conditional probability formula}$$

$$P(G|A \cap B) P(A \cap B) \quad P(A \cap B) = P(B|A)P(A) = 0.75 \cdot 0.3 = 0.225$$

$$0.2 \cdot 0.225 = 0.045$$

$$(b) P(B^c \cap G) = \underset{\text{law of total probability}}{P(G|A \cap B^c) P(A \cap B^c) + P(G|A^c \cap B^c) P(A^c \cap B^c)}$$

$$P(A \cap B^c) = P(B^c|A)P(A) = (1 - 0.75)(0.3) = 0.075$$

$$P(A^c \cap B^c) = \frac{P(B^c | A^c) P(A^c)}{1 - 0.2 \cdot 0.7} = 0.56$$

$$P(B^c \cap G) = 0.8 \cdot 0.075 + 0.9 \cdot 0.56 = 0.564$$

(c) via law of total probability:

$$\begin{aligned} P(G) &= P(G | A \cap B) P(A \cap B) + P(G | A^c \cap B) P(A^c \cap B) + \\ &\quad P(G | A \cap B^c) P(A \cap B^c) + P(G | A^c \cap B^c) P(A^c \cap B^c) \\ &= 0.045 + 0.15 \cdot \underbrace{P(B | A^c)}_{0.2 \cdot (1-0.3)} P(A^c) + 0.075 + 0.56 \\ &= 0.63 \end{aligned}$$

$$\begin{aligned} (d) \quad P(A | G \cap B^c) &= \frac{P(A \cap G \cap B^c)}{P(G \cap B^c)} = \frac{P(G | A \cap B^c) P(A \cap B^c)}{0.564} \\ P(G | A \cap B^c) P(A \cap B^c) &= 0.8 \cdot 0.075 = 0.06 \\ \frac{0.06}{0.564} &\approx 0.1064 \end{aligned}$$